

Updated Values from the Simulation of Radiative Effects with McMule

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McMULE

Expected Cross Sections and Count Rates

Photon Energy	100 MeV	200 MeV	500 MeV
σ_{LO} [mb]	0.250566	0.250566	0.250567
σ_{NLO} [mb]	0.002971	0.002974	0.002978
σ_{inel} [mb]	0.00075	0.00067	0.00057
σ_{tot} [mb]	0.253537	0.253540	0.253545

R_{LO} [1/s]	42.53629	42.53629	42.53652
R_{inel} [1/s]	0.1275	0.1144	0.0973
Photons with $E_\gamma > 2 \text{ GeV}$	55.9 %	62.2 %	73.2 %

Count rate input values:

Target length: 80 cm

Hydrogen pressure: 20 bar

2.687×10^{19} protons/cm³

Beam intensity: 2×10^6 1/s

$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
 $-12. < \theta_\gamma < 12. [\text{mrad}]$

$0.001 < Q^2 < 0.04 [\text{GeV}^2/c^2]$

- LO... Leading order (no photon); NLO... Next to leading order; tot... NLO+LO
- Inel... Bremsstrahlung real photon only
- The real-photon process contributes at the ~0.3% level

McMule vs. ESEPP

	McMule	ESEPP
σ_{el} [mb]	0.25287	0.27901
σ_{inel} [mb]	0.00090	0.00099
σ_{tot} [mb]	0.25376	0.28000

$$0.3 < \theta_{\mu} < 2. [mrad]$$
$$200 \text{ MeV} < E_{\gamma}$$

$$0.001 < Q^2 < 0.04 [GeV^2/c^2]$$

- El... elastic: LO + NLO (without real photon contribution)
- Inel... inelastic: Bremsstrahlung real photon only
- The results differ by ~10%.